

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 3.3 INTRODUCTION TO ELECTRONICS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation. Use your own words to answer the Driving Question: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? Use the evidence from your five-material bulb test, the resistivity data, your class model, the Driving Question Board (DQB) and the Kenyan case studies. Write your answers in the space under each prompt. Use the words conductor, insulator, semiconductor and superconductor carefully, and give numbers and units where you can. Show your thinking in full sentences. Use the data table in Section 2 when it is asked for.

Section 1: What the five-material test showed

Mr Otieno, an electrician in Nairobi, wires a house with copper cable covered in PVC plastic and can safely hold the insulated wire while it carries current at 240 V. In class you tested copper wire, glass, dry wood, plastic and a silicon diode (1N4007) in a 3 V circuit with a torch bulb, always in that order.

- (a) Describe what happened to the bulb for each of the five materials, including both ways of connecting the diode.
- (b) How many of the four ordinary materials (copper, glass, wood, plastic) let the bulb light? What was different about the diode?
- (c) Use your results to explain why Mr Otieno can safely hold the plastic-covered copper cable.

Write your answer here:

Section 2: Using resistivity data to classify materials

The table shows approximate resistivity values at about 20 C.

Material	Resistivity rho (ohm metre)
Copper	1.7×10^{-8}
Germanium (pure)	0.46
Silicon (pure)	2.3×10^3
Dry wood	about 10^8 to 10^{11}
Glass	about 10^{10} to 10^{14}
PVC	about 10^{13} or more

- (a) Put the materials in order from best to worst conductor.
- (b) Sort them into conductor, semiconductor and insulator.
- (c) Calculate how many times more resistive pure silicon is than copper. Show your working.
- (d) How is conductivity related to resistivity?
- (e) Use the data to explain why household wiring uses copper covered in PVC.

Write your answer here:

Section 3: Why materials differ: energy gaps, temperature and doping

All the materials you tested are made of atoms with electrons, yet they behave very differently.

- (a) Using the idea of energy bands and the energy gap (E_g), explain why copper conducts but glass, wood and plastic almost do not. Give the gap values you used.
- (b) Why does pure (intrinsic) silicon conduct a little, and what are the holes that appear? How does warming change its conductivity? How is this different from copper?
- (c) Explain how doping silicon with phosphorus and with boron changes its conduction. Name the type of semiconductor and the majority carriers in each case. Is doped silicon charged?
- (d) Why is doping useful, and how does a p-n junction explain the result you saw with the diode and the bulb?

Write your answer here:

Section 4: Superconductors and MRI machines

Ms Achieng showed a YBCO disc cooled in liquid nitrogen at 77 K (-196 C) making a magnet float. The critical temperatures (T_c) are: mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K.

(a) What is special about the resistance of a superconductor below its T_c , and how is this different from copper?

(b) Dr Wekesa operates an MRI machine at Kenyatta National Hospital. Which coolant (liquid nitrogen at 77 K or liquid helium at 4.2 K) is needed for the NbTi magnet? Explain using the data.

(c) Why does an MRI magnet need a superconductor, and why is household wiring still copper and PVC?

Write your answer here:

Section 5: Final answer to the Driving Question

Write a clear explanation that answers the Driving Question: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines?

Your answer must:

- use at least two pieces of evidence from your tests or data (include numbers and units);
- use the words conductor, insulator, semiconductor and superconductor correctly;
- link to Mr Otieno's wiring and to at least two of these Kenyan cases: the Garissa solar plant (50 MW), microchips in phones, M-Pesa and computers, and the MRI at Kenyatta National Hospital;
- say how your class model changed from Lesson 1 to now.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence and data	Uses accurate results from the bulb test and resistivity, gap and T _c values with correct units and the correct order of materials.	Uses most evidence and numbers correctly, with a few missing units or small slips.	Uses little evidence, or numbers and results are missing or incorrect.
Scientific explanation using the band model	Clearly explains conductors, insulators, semiconductors, temperature effects, doping and p-n junctions using energy gaps, free electrons and holes.	Explains most ideas using gaps and free electrons or holes, with some gaps in reasoning.	Describes what happens but does not explain why, or confuses ideas such as n-type and p-type.

Correct use of terms and conventions	Uses conductor, insulator, semiconductor, superconductor, forward-biased and reverse-biased correctly and says insulators allow almost no current.	Mostly correct terms, with one or two mix-ups.	Often mixes terms or claims insulators allow no current at all.
Connection to Kenyan contexts and the Driving Question	Links the answer to Mr Otieno's wiring and at least two Kenyan cases and answers safety and efficiency fully.	Links to wiring and one other case, answering the Driving Question in part.	Few or no links to the contexts, or the Driving Question is not answered.
Model revision and clarity	Clearly describes how the class model changed from Lesson 1 and writes in well-organised sentences.	Mentions model change and writes clearly most of the time.	Does not describe model change, or writing is hard to follow.

ATTRIBUTION AND LICENSING

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